SIDE 1

STAT 251 · EXAM 2 — Calculation Sheet

A Which procedure? · 1–4 Module 4 · 5–6 z / t tables · 7 SE · Welch (side 2) · 8, 10 CI (side 2) · 11–13 Tests (side 2) · 14 Worked (side 2) · 15–16 MC (side 2)

A · If the question says…	Parameter → estimate	Two-sided CI (estimate ± crit × SE)	Test statistic (under H_0)	Table · df (conditions → §13)
"% / proportion of all…", one yes/no group	$p \to \hat{p} = \frac{x}{n}$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$ ← uses p_0!	**z**
compare % of two separate groups	$p_1 - p_2 \to \hat{p}_1 - \hat{p}_2$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$	$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$ (lecture: unpooled)	**z**
"average / mean of all…", one group, s given	$\mu \to \bar{y}$ $(= \bar{x})$	$\bar{y} \pm t^*_{n-1} \frac{s}{\sqrt{n}}$	$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$	**t**, $df = n - 1$
compare means of two separate groups	$\mu_1 - \mu_2 \to \bar{y}_1 - \bar{y}_2$	$(\bar{y}_1 - \bar{y}_2) \pm t^*_\nu \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$t = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$, Δ_0 usually 0	**t**, Welch ν (§7, side 2) — round **down**
same subjects before/after, matched pairs	$\mu_d \to \bar{d}$ (one list d_i)	$\bar{d} \pm t^*_{n-1} \frac{s_d}{\sqrt{n}}$, n = # pairs	$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}$, d_0 usually 0	**t**, $df = n - 1$
Normal X: P(value…) with μ, σ known	—	reverse: $x = \mu + z\sigma$	$z = \frac{x - \mu}{\sigma}$	**z**
P(sample mean …) with μ, σ known	—	$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$	$z = \frac{\bar{y} - \mu}{\sigma/\sqrt{n}}$	**z**
"how many to sample for margin E?"	proportion	$n = \left\lceil \frac{(z^*)^2 \hat{p}(1-\hat{p})}{E^2} \right\rceil$ — no prior estimate → $\hat{p} = .5$; **always round UP**; E as a decimal (3% = .03)		**z***

POPULATION (parameters & claims)

$\boldsymbol{p}$ = true population proportion (unknown). *"the true % / proportion of all…"*
$\boldsymbol{\mu}$ = true population mean (unknown). *"true average of all…", "mean of the population"*
$\boldsymbol{\sigma}$ = population SD; $\boldsymbol{\sigma^2}$ = variance. *"population SD is…", "SD = 0.04 s"; known $\sigma \to z$*
$\boldsymbol{p_0}$, $\boldsymbol{\mu_0}$ = value claimed in $\boldsymbol{H_0}$ (null value). *"claims 52%", "is the mean 0.400?"*
$\boldsymbol{q_0 = 1 - p_0}$ = complement of the null proportion.
$\boldsymbol{p_1 - p_2}$, $\boldsymbol{\mu_1 - \mu_2}$ = difference between groups 1 and 2. *"compare A vs B"*
$\boldsymbol{\mu_d}$ = true mean of paired differences. *"before/after on the same people"*
$\boldsymbol{\Delta_0}$, $\boldsymbol{d_0}$ = claimed difference in $\boldsymbol{H_0}$ (usually 0). *"no difference"*
$\boldsymbol{N}$ = population size (10% condition).

SAMPLE (what the data give you)

$\boldsymbol{n}$ = sample size (# in sample; paired: # of pairs). *"randomly sampled 36…"*
$\boldsymbol{x}$ = # of "successes" (yes answers). *"62 of 100 prefer…"*
$\boldsymbol{\hat{p} = \frac{x}{n}}$ = sample proportion ("p-hat"). *"62% of those sampled…"*
$\boldsymbol{\hat{q} = 1 - \hat{p}}$ = sample proportion of "no".
$\boldsymbol{\bar{y}}$ $\boldsymbol{(= \bar{x})}$ = sample mean ("y-bar"). *"the sample averaged 0.424 s"*
$\boldsymbol{s}$ = sample SD; $\boldsymbol{s^2}$ = sample variance. *"sample standard deviation 0.060" → t*
$\boldsymbol{n_1, \bar{y}_1, s_1, \hat{p}_1}$ = values for group 1 (subscript 2 = group 2).
$\boldsymbol{d_i}$ = one pair's difference (e.g. after − before).
$\boldsymbol{\bar{d}}$, $\boldsymbol{s_d}$ = mean and SD of the d_i list.

INFERENCE TOOLS

PE = point estimate (best single guess: $\hat{p}$, $\bar{y}$, …).
$\boldsymbol{SE}$ = standard error = SD of the estimate.
$\boldsymbol{z^*}$, $\boldsymbol{t^*_{df}}$ = critical value from the confidence level. *"95% confident", "90% CI"*
$\boldsymbol{E}$ (ME, $\boldsymbol{m}$) = margin of error $= \boldsymbol{c^* \times SE}$. *"within ±3 points", "accurate to .01"*
$\boldsymbol{C}$ = confidence level; $\boldsymbol{\alpha = 1 - C}$.
$\boldsymbol{\alpha}$ = significance level (P of Type I error). *"at the 5% level", "α = .05"*
$\boldsymbol{df}$, $\boldsymbol{\nu}$ = degrees of freedom (row of the t-table).
$\boldsymbol{z}$, $\boldsymbol{t}$ = test statistic (# of SEs from the null).
p-value = P(result this extreme or more | $\boldsymbol{H_0}$ true).
$\boldsymbol{H_0}$, $\boldsymbol{H_a}$ = null (has =) / alternative ($<$, $>$, $\neq$). *"test whether…", "is there evidence…"*
$\boldsymbol{\beta}$ = P(Type II); power $\boldsymbol{= 1 - \beta}$.

MODULE 4 SYMBOLS

$\boldsymbol{X}$ = the random variable (the quantity measured). *"let X = time until…"*
$\boldsymbol{x}$ = one specific value of $\boldsymbol{X}$.
$\boldsymbol{f(x)}$ = PDF (density curve; area = probability). *"the pdf is…"*
$\boldsymbol{F(x)}$ = CDF $= \boldsymbol{P(X \le x)}$, area to the left. *"the cdf", "at most x"*
$\boldsymbol{E[X]}$, $\boldsymbol{\mathrm{Var}(X)}$ = mean / variance of $\boldsymbol{X}$. *"expected value", "average"*
$\boldsymbol{a}$, $\boldsymbol{b}$ = Uniform min and max. *"anywhere between a and b"*
$\boldsymbol{c}$, $\boldsymbol{d}$ = endpoints of the interval asked about.
$\boldsymbol{\lambda}$ = rate (events per time unit). *"5 per hour", "on average 2 per hour"*
$\boldsymbol{k}$ = unknown constant in a PDF. *"find k so f is a valid pdf"*
$\boldsymbol{Z}$, $\boldsymbol{\Phi(z)}$ = standard Normal; table area left of $\boldsymbol{z}$.
$\boldsymbol{x_p}$, $\boldsymbol{m}$ = $\boldsymbol{p}$-th percentile; median ($\boldsymbol{F(m) = .5}$). *"90th percentile", "median"*

1 Continuous RVs: PDF / CDF toolkit

$$f(x) \ge 0 \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(X = x) = 0$$

$$P(a < X < b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt \qquad f(x) = F'(x)$$

$P(X > x) = 1 - F(x)$ $\quad <$ and $\le$ give the **same** answer

$$\mu = E[X] = \int x\,f(x)\,dx \qquad E[X^2] = \int x^2 f(x)\,dx$$

$$\sigma^2 = \mathrm{Var}(X) = E[X^2] - \mu^2 = \int (x-\mu)^2 f(x)\,dx, \qquad \sigma = \sqrt{\sigma^2}$$

Typical asks → do this:

- **find** k: set $\int_{\text{support}} f(x)\,dx = 1$, solve.
- **CDF from PDF**: $F(x) = \int_{\text{low}}^{x} f(t)dt$ on the support; $F = 0$ below, $F = 1$ above (write it piecewise).
- **median** m: solve $F(m) = 0.5$. **p-th percentile**: $F(x_p) = p$.
- "at most / below x" → $F(x)$; "more than" → $1 - F(x)$; "between" → $F(b) - F(a)$.

Integrals: $\int x^n dx = \frac{x^{n+1}}{n+1}$, $\int e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x}$.

Practice template $f(x) = kx$, $0 \le x \le 2$: $\int_0^2 kx\,dx = 2k = 1 \Rightarrow k = \frac{1}{2}$. $F(x) = \frac{x^2}{4}$. $P(X < 1) = \frac{1}{4}$. $E[X] = \int_0^2 \frac{x^2}{2} dx = \frac{4}{3}$, $E[X^2] = \int_0^2 \frac{x^3}{2} dx = 2$, $\mathrm{Var} = 2 - \frac{16}{9} = \frac{2}{9}$. Median: $\frac{m^2}{4} = .5 \Rightarrow m = \sqrt{2} \approx 1.414$.

2 Uniform $X \sim U(a, b)$

If you see: "equally likely anywhere between a and b", flat density.

$$f(x) = \frac{1}{b-a},\ a \le x \le b \qquad F(x) = \frac{x-a}{b-a}$$

$$P(c < X < d) = \frac{d-c}{b-a} \qquad E[X] = \frac{a+b}{2} \qquad \mathrm{Var} = \frac{(b-a)^2}{12}$$

where a = smallest possible value · b = largest · c, d = interval asked about · $b - a$ = total width · p = percentile fraction (.90 for 90th).

SOCCER drill starts $U(0, 20)$ min: $P(5 < X < 10) = \frac{5}{20} = .25$; mean 10; percentile: $x_p = a + p(b - a)$.

3 Exponential $X \sim \mathrm{Exp}(\lambda)$

If you see: waiting time until the *first/next* event; λ = **rate per time unit**.

$$f(x) = \lambda e^{-\lambda x},\ x \ge 0 \qquad F(x) = 1 - e^{-\lambda x} \qquad P(X > x) = e^{-\lambda x}$$

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b} \qquad E[X] = \sigma = \frac{1}{\lambda},\ \mathrm{Var} = \frac{1}{\lambda^2}$$

$$\text{percentile: } x_p = \frac{-\ln(1-p)}{\lambda} \qquad \text{median} = \frac{\ln 2}{\lambda}$$

$$\text{memoryless: } P(X > s + t \mid X > s) = P(X > t)$$

where λ = average # of events per time unit · x = waiting time (same units as λ!) · $e \approx 2.718$ · $\ln$ = natural log · p = percentile fraction · s, t = time already waited / extra time.

LECTURE $\lambda = 5$/hour, first customer between 15 and 60 min. **Convert units first**: 0.25 h, 1 h.
$P(0.25 < X < 1) = e^{-5(0.25)} - e^{-5(1)} = .2865 - .0067 \approx$ **.2798**
Rate 5/hour ≠ wait of 5 hours. 2 goals/hour → mean wait $\frac{1}{2}$ h = 30 min.

4 Normal $X \sim N(\mu, \sigma^2)$

If you see: bell-shaped / Normal, μ and SD given. **Trap:** $N(\mu, \sigma^2)$ — 2nd number is the **variance**; $\sigma = \sqrt{\sigma^2}$.

$$\text{standardize: } z = \frac{x - \mu}{\sigma} \qquad \text{un-standardize: } x = \mu + z\sigma$$

where x = raw value (e.g. 0.46 s) · μ = population mean · σ = population SD · z = # of SDs above (+) / below (−) the mean.

Probability: $x \to z \to$ table area (§5). **Find a value/percentile:** area → find it *inside* the table → $z \to x = \mu + z\sigma$ (top 10% → left area .90 → $z = 1.28$).

68–95–99.7: within $1\sigma, 2\sigma, 3\sigma$. SOCCER reaction $N(.40, .04^2)$: 68% in .36–.44 s; 95% in .32–.48 s (approx. only — use the table for exact).

5 z-table: area to the LEFT of z

$$\text{table entry} = \Phi(z) = P(Z \le z)$$

where Z = standard Normal (mean 0, SD 1) · $\Phi(z)$ = area LEFT of z.

Row = z to 1 decimal, column = 2nd decimal. $z = 1.96$: row 1.9, col .06 → .9750. Negative z → negative half of the table.

Want	Do	e.g. $z = 1.20$ ($\Phi = .8849$)
left $P(Z < z)$	table	.8849
right $P(Z > z)$	$1 - \Phi(z)$	.1151
between a, b	$\Phi(b) - \Phi(a)$	bigger − smaller
outside $\pm\lvert z\rvert$	$2[1 - \Phi(\lvert z\rvert)]$	$z = 2$: $2(.0228) = .0456$
z from an area	find area inside table	.8849 → 1.20

z^* **(two-sided, look up left area $1 - \frac{\alpha}{2}$):** 80% 1.282 · 90% 1.645 · **95% 1.960** · 98% 2.326 · 99% 2.576. **One-sided** (α in one tail): .10 → 1.282 · .05 → 1.645 · .01 → 2.326.

6 t-table: critical t* for an UPPER tail

Rows = **df**, columns = upper-tail probability $P(T > t^*)$. It does **not** give cumulative areas. Means use t because σ is unknown (we use s) → heavier tails; large df → $t \approx z$.

Pick the column: two-sided CI at C → column $\alpha/2$ (95% → .025; 90% → .05; 99% → .005). One-sided bound/test → column α.
Pick the row: one mean / paired: $n - 1$. Two means: Welch ν, **round DOWN** (e.g. 22.38 → 22). Row missing (e.g. 35 when the table jumps 30→40)? Use the **lower** df row (conservative).

p-value by bracketing: in your df row find the two critical values around $|t|$ → p is between their column headings (one-tail). **Double both** for H_a :≠. Bigger than every entry → p < smallest heading.
e.g. $df = 35$: $2.030 < \mathbf{2.40} < 2.438$ → $.01 < p < .025$ (one-tail).

★ Module 4 time-savers

- **Unit change** $Y = aX + b$: $E[Y] = aE[X] + b$, $\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$.
- **Exponential backwards:** $P(X > x) = q \Rightarrow x = \frac{-\ln q}{\lambda}$; mean given → $\lambda = \frac{1}{\text{mean}}$.
- **Normal:** $P(-z < Z < z) = 2\Phi(z) - 1$; solve $\sigma = \frac{x-\mu}{z}$ or $\mu = x - z\sigma$.

Inference: CIs · Tests · Worked Examples · MC

8 CI master rule

$$\text{CI} = \text{PE} \pm \underbrace{c^* \times SE}_{\text{margin of error } E}$$

1. Parameter? → PE + SE formula (table A, side 1).
2. Critical value: proportions z^* (§5), means t^* (§6).
3. $E = c^* \cdot SE$; interval $(\text{PE} - E,\ \text{PE} + E)$.
4. Conditions (§13) + interpretation sentence.

Interpretation template: "We are 95% confident that the true *[parameter in context]* is between ___ and ___."
Never: "95% probability the parameter is in this interval." Long-run meaning: 95% of intervals built this way capture the parameter.

Reading a given CI (L, U): $\text{PE} = \frac{L+U}{2}$, $E = \frac{U-L}{2}$, $SE = E/c^*$, width $= 2E$.
Width ↑ when confidence ↑ or n ↓ or variability ↑. Width $\propto 1/\sqrt{n}$ (×4 n → half width).

One-sided bounds ($\alpha = 1 - C$ in ONE tail)

lower: $(\text{PE} - c^*SE,\ \infty)$ upper: $(-\infty,\ \text{PE} + c^*SE)$

c^*: z with left area C (90% → 1.282), or t from column α (90% → .10 column). Professor's equivalent: negative z with left area α, compute PE + z·SE.
All five CI formulas are in table A (side 1). t^*_{n-1}: subscript = df, not "t* minus 1". Paired: n = # pairs.

Difference CIs: look at 0
Contains 0 → "no difference" plausible · all > 0 → group 1 larger · all < 0 → group 1 smaller.

10 Sample size for a proportion

$$E = z^*\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \implies n = \left\lceil \frac{(z^*)^2\,\hat{p}(1-\hat{p})}{E^2} \right\rceil$$

Use a prior $\hat{p}$ if given, else .5 ($.5 \times .5 = .25$ is the max → safest). **Round UP**, even 9603.1 → 9604.

E ($\hat{p}$ = .5)	.01	.02	.03	.04	.05
90% (1.645)	6766	1692	752	423	271
95% (1.96)	**9604**	2401	1068	601	385
99% (2.576)	16590	4148	1844	1037	664

Halve E → 4× the sample. $\frac{1.96^2(.5)(.5)}{.01^2} = 9604$.

Reverse z lookups (percentiles & cutoffs)

Want	left area to find	z
top 10% / 90th pct	.9000	1.28
top 5% / 95th pct	.9500	1.645
top 2.5%	.9750	1.96
top 1%	.9900	2.33
bottom 10% / 5%	.1000 / .0500	−1.28 / −1.645
middle 90% / 95% / 99%	tails .05 / .025 / .005	±1.645 / ±1.96 / ±2.576
quartiles Q1 / Q3	.2500 / .7500	−0.67 / 0.67

Then $x = \mu + z\sigma$ (individual) or $\bar{y} = \mu + z\,\sigma/\sqrt{n}$ (sample mean). Area between two closest table entries → pick the closer one (or average).

7 Sampling distributions, SE & Welch df

$$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right),\ SE(\bar{y}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right),\ SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

where $\bar{Y}$ = sample mean (random) · c = cutoff in the question · other symbols: key on side 1.
SD = spread of individuals; **SE** = spread of a statistic over repeated samples. n ×4 → SE ÷2.

CLT probability: $P(\bar{Y} > c) = 1 - \Phi\left(\frac{c-\mu}{\sigma/\sqrt{n}}\right)$. Practice: $\mu = .40, \sigma = .04, n = 16$: SE = .01; $P(\bar{Y} > .42) = 1 - \Phi(2.00) = .0228$.

Sample SD (e.g. of paired d_i): $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}\ \text{("crazy df")}$$

where s_1, s_2 = sample SDs · n_1, n_2 = group sizes · x_i = each data value · $\bar{x}$ = their mean · ν ("nu") = Welch df.
Compute $A = s_1^2/n_1$, $B = s_2^2/n_2$ first: $\nu = (A+B)^2/\left(\frac{A^2}{n_1-1} + \frac{B^2}{n_2-1}\right)$. If allowed, conservative shortcut $df = \min(n_1, n_2) - 1$.

11 Hypothesis test workflow

1. H_0 (has "=", the claim/default) and H_a (<, >, ≠) — choose before seeing data.
2. Check conditions (§13).
3. Test statistic = $\frac{\text{estimate} - \text{null value}}{SE \text{ under } H_0}$ (table A).
4. Tail from H_a → p-value from the right table.
5. $p \le \alpha$ → **reject** H_0; $p > \alpha$ → fail to reject H_0.
6. Conclusion in context.

Wording → H_a: "less / decreased / lower" → < · "more / increased / slower(time ↑)" → > · "different / changed / not equal" → ≠.
Conclusion template: "Since p = ___ (< or >) α = ___, we (reject / fail to reject) H_0. There is (enough / not enough) evidence that *[H_a in context]*."

12 p-value: tail × table

H_a	z-test (table = left area)	t-test (bracket in df row)
< left	$p = \Phi(z)$	one-tail area beyond $\lvert t\rvert$
> right	$p = 1 - \Phi(z)$	one-tail area beyond $\lvert t\rvert$
≠ two	$p = 2[1 - \Phi(\lvert z\rvert)]$	2 × one-tail bracket

e.g. $z = -1.20$: H_a: <→ .1151; H_a: >→ .8849; H_a: ≠→ .2302.

Rejection regions (critical-value method): $\alpha = .05$ — left: $z < -1.645$ · right: $z > 1.645$ · two: $|z| > 1.96$. For t: same idea with t^* from the df row (column α one-sided, $\alpha/2$ two-sided).
CI ↔ test: two-sided test at α rejects $H_0: \theta = \theta_0$ exactly when θ_0 is outside the $(1-\alpha)$ CI.

13 Conditions (write them!)

	Random	10%	Shape / counts	Indep. groups
1 prop	✓	✓	CI $n\hat{p}, n\hat{q} \ge 10$; test $np_0, nq_0 \ge 10$	—
2 props	both	both	S/F for both	✓
1 mean	✓	✓	Normal-ish; skew → $n > 30$; heavy skew → $n > 100$	—
2 means	both	both	same, both groups	✓
paired	✓	✓	differences Normal-ish	pairs dependent!

10%: $n \le 0.10N$ (e.g. $n = 36$ needs population ≥ 360).

14 Worked examples

LECTURE One proportion test (voters): $H_0: p = .52$, $H_a: p < .52$, $n = 10{,}000$, $\hat{p} = .514$.

$$z = \frac{.514 - .52}{\sqrt{.52(.48)/10000}} = \frac{-.006}{.004996} \approx -1.20$$

left tail: $p = \Phi(-1.20) = .1151$

$.1151 > .05$ → fail to reject; not enough evidence support is below 52%. CI version (uses $\hat{p}$): $.514 \pm 1.96(.004998)$ → (50.42%, 52.38%).

LECTURE Two proportions (cats): F 13/45 = .2889, M 16/74 = .2162, $H_a: p_F - p_M > 0$, unpooled SE:

$$z = \frac{.2889 - .2162}{\sqrt{\frac{.2889(.7111)}{45} + \frac{.2162(.7838)}{74}}} = \frac{.0727}{.0828} \approx 0.878$$

right tail: $p = 1 - \Phi(0.88) = 1 - .8106 = .1894 > .10$ → fail to reject (insufficient evidence).

LECTURE One mean (HS GPA): $H_0: \mu = 4.1$, $H_a: \mu \ne 4.1$, $n = 114$, $\bar{y} = 4.2317$, $s = .3664$:

$$t = \frac{4.2317 - 4.1}{.3664/\sqrt{114}} = \frac{.1317}{.03432} \approx 3.84, \quad df = 113$$

Row 113 missing → use lower row (e.g. 100). t beyond the last column (one-tail .001) → two-sided $p < .002$ (lecture: $p < .001$; exact .0002). $\alpha = .02$ → reject: mean differs from 4.1.

LECTURE Two means (1st-gen vs not): $n_1 = 17, \bar{y}_1 = 4.1467, s_1 = .3495$; $n_2 = 102, \bar{y}_2 = 4.2466, s_2 = .369$; $H_a: \mu_1 - \mu_2 < 0$.

$$t = \frac{4.1467 - 4.2466}{\sqrt{\frac{.3495^2}{17} + \frac{.369^2}{102}}} = \frac{-.0999}{.0923} \approx -1.083$$

$\nu \approx 22.38 \to 22$, left tail: $p \approx .145$

$.145 > .03$ → fail to reject.

SOCCER One-mean t-test: claim $\mu = .400$ s, suspect slower: $H_a: \mu > .400$; $n = 36$, $\bar{y} = .424$, $s = .060$, $\alpha = .05$.

$$t = \frac{.424 - .400}{.060/\sqrt{36}} = \frac{.024}{.010} = 2.40,\ df = 35$$

Row 35: $2.030 < 2.40 < 2.438$ → $.01 < p < .025 < .05$ → reject; mean reaction time exceeds .400 s. (Row 30 instead: 2.042 < 2.40 < 2.457, same result.)

SOCCER One-proportion CI: 62 of 100 prefer morning training, 95%:

$$SE = \sqrt{\frac{.62(.38)}{100}} = .04854, \quad E = 1.96(.04854) = .0951$$

$$.62 \pm .0951 = (.5249,\ .7151)$$

S/F: 62, 38 ≥ 10 ✓. Testing $H_0: p = .50$ instead → $SE = \sqrt{.5(.5)/100} = .05$, $z = \frac{.62-.50}{.05} = 2.40$.

SOCCER Paired recipe: each keeper's d_i = after − before → $\bar{d}$, s_d (§7 formula) → $t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$, df = pairs − 1.
Never use the two-sample formula on paired data.

15 MC must-knows

- **Parameter** (p, μ; population, unknown) vs **statistic** ($\hat{p}, \bar{y}, s$; sample).
- **p-value** = P(data this extreme or more | H_0 true). It is NOT P(H_0 true).
- Fail to reject ≠ "accept H_0" / ≠ proof H_0 is true.
- **Type I**: reject a true H_0 (false alarm), prob $= \alpha$. **Type II**: fail to reject a false H_0 (miss), prob $= \beta$. **Power** $= 1 - \beta$ = P(correctly reject false H_0).
- Power ↑ with larger n, larger effect size, larger α (but Type I risk ↑). ↓α → ↑β.
- Confidence level ↑ → wider CI; n ↑ → narrower CI.
- t vs z: t heavier tails, depends on df; df → ∞ gives z.
- CI uses $\hat{p}$ in SE; test uses p_0 in SE.
- Continuous: $P(X = x) = 0$; probability = area; density can exceed 1 (area can't).
- Uniform flat · Exponential = time until first event (constant rate) · Normal symmetric bell.
- H_a is chosen before seeing data; tail comes only from H_a.

16 Trap list

- p (proportion) ≠ p-value · Var vs SD in $N(\mu, \sigma^2)$ · minutes vs hours for λ.
- z-table is LEFT area; t-table is UPPER-tail critical values — don't read t like z.
- Two-sided: split α (.025 each tail) for CIs; double one-tail p for ≠ tests.
- Paired ≠ two independent groups. Means → t, proportions → z.
- Sample size: round **up**; E as decimal.
- Zero in a difference CI = no difference plausible.

Two-group CI recipes (step by step)

$p_1 - p_2$: ① $\hat{p}_1 = \frac{x_1}{n_1}$, $\hat{p}_2 = \frac{x_2}{n_2}$ ② PE = $\hat{p}_1 - \hat{p}_2$ ③ $SE = \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$ ④ z^* ⑤ PE $\pm z^* SE$ ⑥ check $n_1\hat{p}_1, n_1\hat{q}_1, n_2\hat{p}_2, n_2\hat{q}_2 \ge 10$ ⑦ read vs 0.

$\mu_1 - \mu_2$: ① $A = \frac{s_1^2}{n_1}$, $B = \frac{s_2^2}{n_2}$ ② $SE = \sqrt{A + B}$ ③ $\nu = \frac{(A+B)^2}{A^2/(n_1-1) + B^2/(n_2-1)}$, round down ④ t^* from row ν, column $\alpha/2$ ⑤ $(\bar{y}_1 - \bar{y}_2) \pm t^* SE$ ⑥ read vs 0.

Paired: ① d_i = (after − before) for every subject ② $\bar{d} = \frac{\sum d_i}{n}$ ③ $s_d = \sqrt{\frac{\sum(d_i - \bar{d})^2}{n-1}}$ ④ t^*_{n-1} ⑤ $\bar{d} \pm t^* \frac{s_d}{\sqrt{n}}$.

t* t-table check strip (upper-tail areas)

df	.10	.05	.025	.01	.005
10	1.372	1.812	2.228	2.764	3.169
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
∞ (z)	1.282	1.645	1.960	2.326	2.576

Only to confirm your table reading; t* → z* as df grows.

Sentence bank (numbers = placeholders)

- **CI:** "We are 95% confident the true mean reaction time of all college goalkeepers is between .405 and .443 s."
- **Difference CI:** "...true difference $p_1 - p_2$ is between −.04 and .12; since 0 is inside, no difference is plausible."
- **Reject:** "p = .015 ≤ α = .05, so reject H_0. There is convincing evidence that [H_a in context]."
- **Fail to reject:** "p = .19 > α = .10, so fail to reject H_0. There is not enough evidence that [H_a in context]."